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碩士學位論文

以資訊瓶頸原理強化聯邦學習之個性化隱私保護能力

Personalized Privacy Preservation in Federated Learning via the Information Bottleneck Principle

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摘　要

聯邦學習 (FL) 已成爲隱私敏感環境下協作機器學習的一項頗具前景的範例，它允許多個客户端在不直接共享原始資料的情況下聯合訓練全局模型。然而，儘管聯邦學習系統本身俱有隱私優勢，但最近的研究表明，它仍然容易受到成員推理攻擊 (MIA) 或還原攻擊等威脅影響。爲了解決這個問題，我們提出了一個基於資訊瓶頸 (IB) 的聯邦學習框架，有效地降低了這些攻擊帶來的風險。透過在每個客户端引入可調的壓縮參數，我們的方法能夠靈活地控制本地表示中保存的資訊量，使模型學習到的表示包含的輸入的資訊量盡可能減少，從而允許每個客户端根據自身需求在模型效用和隱私保護之間取得平衡。在實驗結果我們通過 MIA 攻擊驗證了我們方法對隱私保護的有效性，同時模型也能有效的維持效能，尤其是在資料分佈高度非獨立同分佈 (Non-IID) 的場景下。這些發現表明，我們的方法是一種實用有效的解決方案，可用於增強聯邦學習的隱私保護。

關鍵字— 聯邦學習, 資訊瓶頸理論, 成員推斷攻擊, 非獨立同分佈數據

Abstract

Federated Learning (FL) has emerged as a promising paradigm for collaborative machine learning in privacy-sensitive environments, enabling multiple clients to jointly train a global model without directly sharing their raw data. However, despite its inherent privacy advantages, recent studies have shown that FL remains vulnerable to threats such as Membership Inference Attacks (MIA) and reconstruction attacks. To address this issue, we propose an information bottleneck (IB)-based federated learning framework, which effectively mitigates these risks. By introducing an adjustable compression parameter at each client, our approach flexibly controls the amount of information retained in local representations, encouraging the learned features to minimize unnecessary input information. This design allows each client to balance model utility and privacy protection according to their specific requirements. Experimental results demonstrate the effectiveness of our defense against MIA, while maintaining competitive model performance, particularly under highly Non-IID data distributions. These findings suggest that our method provides a practical and effective solution for enhancing privacy protection in Federated Learning systems that are vulnerable to membership inference attacks.

Keywords— Federated Learning, Information Bottleneck, Membership Inference Attack, Non-IID data

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Chapter 1 Introduction

In recent years, federated learning (FL) [1] has become a cornerstone technology for privacy-aware collaborative machine learning, empowering various organizations and devices to contribute to model training without directly exchanging raw data. This decentralized approach is particularly valuable in domains where data privacy and regulatory compliance are paramount, such as healthcare, finance, and smart devices. FL operates by aggregating locally computed updates—such as synthetic data or model parameters—from participating nodes, thereby mitigating many privacy risks associated with centralized data collection.

Nevertheless, federated learning is not immune to privacy threats. Advanced inference attacks, including membership inference attack [2–4] and reconstruction attacks [5,6]. For example, if multiple hospitals collaboratively train a disease recognition model, an adversary employing MIA could infer whether a particular individual has visited a specific hospital, thereby compromising patient confidentiality. The effectiveness of MIA underscores the importance of developing robust privacy-preserving mechanisms within FL systems.

Numerous privacy-preserving techniques have been proposed in the context of FL, such as differential privacy (DP) [7] and sparsification [8]. These approaches typically operate by perturbing the uploaded gradients or model parameters through noise injection or sparsification, which effectively strengthens privacy guarantees by impeding potential adversaries. However, such defensive mechanisms inevitably interfere with global model aggregation, often resulting in degraded overall performance.

Furthermore, most existing privacy defense strategies adopt a uniform protection mechanism across all clients, without sufficiently considering the heterogeneity of local data distributions or the varying privacy risks encountered by individual clients. In practical FL scenarios, data distributions among clients are often highly non-IID, and the requirements and risk profiles for privacy protection can vary significantly across participants. Applying the most stringent privacy-preserving measures indiscriminately to all clients may enhance overall security, but typically leads to excessive degradation in model performance due to the mismatch between the level of protection and the actual needs or risks faced by each client.

To address these issues, we draw inspiration from the Information Bottleneck (IB) principle, which demonstrates strong potential for feature compression. We propose a privacy-preserving framework for federated learning based on the IB paradigm. The information bottleneck framework, rooted in information theory [9], is employed to compress client-specific privacy features for effective privacy protection. By using IB to preserve only the information most relevant to the target task and discard redundant or sensitive details, our approach removes privacy attributes unrelated to the learning objective. This systematic restriction of mutual information substantially reduces the risk of privacy leakage in federated learning environments.

To accommodate the diverse privacy requirements of clients, our approach introduces a dynamic defense adaptation mechanism. Specifically, we assess the privacy risk of each client by analyzing the distribution of its local data. Based on this assessment, we adaptively adjust the strength of privacy protection for each client through the compression parameter in the information bottleneck framework. Clients identified as having higher privacy risk are assigned stronger compression, while those with lower risk can operate with reduced protection, thereby maximizing overall model accuracy. This strategy ensures that privacy is preserved where it is most needed, without unnecessarily sacrificing global model utility. Consequently, our framework enables differentiated privacy protection, allowing each client to balance privacy and model accuracy according to its unique risk profile.

Comprehensive experimental results further demonstrate that the proposed method not only provides robust defense against inference attacks across a wide range of non-IID data distributions, but also maintains competitive model performance. These findings underscore the practicality and effectiveness of supporting flexible, client-specific privacy configurations in federated learning systems, rather than imposing a uniform and potentially over-conservative privacy standard on all participants. The contributions of this paper are summarized as follows:

- We propose an information bottleneck-based federated learning framework that flexibly balances privacy protection and model utility via adaptive compression.
- Our method supports client-level personalized privacy guarantees, allowing each client to select the appropriate privacy-utility trade-off based on local requirements.
- Extensive experiments demonstrate that our approach achieves superior privacy protection and competitive model performance, particularly under challenging non-IID scenarios.

Chapter 2 Related Work

2.1 Security Concerns in Federated Learning

In Federated Learning, privacy threats arise across the entire pipeline, including the server, clients, and their communication. The server might be compromised by malicious updates or targeted attacks, while clients often possess sensitive or noisy data that, if inadequately protected, could expose private information. These vulnerabilities are exacerbated by sophisticated attacks that exploit the collaborative nature of FL to extract or infer sensitive training data.

Model inversion attacks exploit gradients or model parameters shared during training to reconstruct private user data, posing a significant threat to privacy in FL. Zhu et al. [5] proposed iDLG, which reconstructs individual training samples by iteratively matching generated gradients to the observed gradients, effectively recovering sensitive inputs. Geiping et al. [6] extended this concept with a gradient inversion technique capable of recovering high-fidelity images even from aggregated gradients, highlighting how gradient information inherently encodes detailed properties of the original data.

In contrast, Membership Inference Attacks target a different privacy dimension by inferring whether a particular data point was part of the training dataset. Early work by Shokri et al. [2] introduced the shadow training technique to construct attack models that distinguish between member and non-member samples, demonstrating the feasibility of such attacks even against well-generalized models. In the FL setting, Gu et al. [10] proposed CS-MIA, which leverages changes in prediction confidence over multiple training rounds to achieve high membership inference accuracy. Unlike inversion attacks, MIAs are often less affected by conventional defenses such as pruning or differential privacy, as these defenses primarily perturb gradients or limit individual data contributions but do not necessarily eliminate membership signals embedded in the learned representations. This resilience makes MIAs particularly concerning for privacy-sensitive applications deployed under the federated paradigm.

2.2 MIA in Federated Learning

Membership inference attacks have received growing attention in the context of Federated Learning, where shared gradients or model updates may inadvertently leak membership information. Nasr et al. [4] conducted one of the earliest systematic analyses of MIA in FL, demonstrating how traces embedded in the gradients of the loss function could be exploited to infer whether a specific data point participated in training. Building on this line of inquiry, Li et al. [11] introduced a passive membership inference attack that operates without the need for training on explicit member samples, thereby lowering the attacker's prior knowledge requirements. More recently, Zhu et al. [12] proposed FedMIA, which leverages estimation of the underlying data distribution to carry out membership inference, further highlighting how statistical properties extracted from federated updates can compromise user privacy. These studies collectively underscore the vulnerability of FL frameworks to MIAs, motivating the need for more robust privacy-preserving mechanisms.

2.3 Defense Against MIA

Data augmentation techniques have been explored as a means to implicitly enhance privacy by broadening and smoothing the underlying data distribution. Approaches such as MixUp [13] and InstaHide [14] adopt this paradigm by linearly interpolating pairs of training samples to generate synthetic examples, thereby making it more challenging for adversaries to isolate individual data points. These methods aim to obscure membership signals or fine-grained patterns that could be exploited by attacks like membership inference, offering a lightweight strategy that complements more explicit privacy-preserving mechanisms. Another line of research focuses on directly mitigating privacy leakage through manipulating the gradients exchanged during federated learning. Gupta et al. [8] proposed transmitting only sparsified subsets of gradients to reduce the information exposed in each communication round. Differential Privacy remains one of the most widely studied techniques in this category, where calibrated noise is added to gradients prior to upload. For instance, Shokri et al. [7] employed DP in an FL setting by having each participant inject Laplacian noise into their updates. However, their approach suffered from significant accuracy degradation and imposed heavy computational overhead, since privacy budgets were accounted for on a per-parameter basis—rendering it less practical for deep neural networks with large parameter spaces. To address these limitations, Yang et al. [15] designed a perturbation strategy that minimizes impact on model utility. More recently, Liu et al. [16] introduced a novel defense termed Poisoning as a Post-Protection, which injects carefully crafted perturbations after training to further obscure sensitive information embedded in gradients, thereby enhancing resilience against inference attacks.

Nevertheless, these studies do not consider the varying levels of risk arising from differences in local data distributions among clients. To address privacy requirements stemming from heterogeneity in client data distributions, recent research has investigated personalized privacy protection mechanisms. Early works achieved flexible privacy-utility trade-offs by allocating differentiated privacy budgets to individual clients [17]. Subsequent studies expanded upon this direction by tailoring privacy parameters or budgets according to user preferences, data contributions, or contextual dynamics, thus enabling adaptive and customized protection for each client [18,19]. More recent research, such as FedDPA [20], has further explored risk-aware differential privacy schemes, where privacy levels are determined based on privacy risks estimated from each client's data distribution.However, all of the aforementioned personalized privacy studies rely on DP-based mechanisms, which are susceptible to performance degradation as model complexity increases.

2.4 IB with Federated Learning

In the realm of Federated Learning, various studies have explored leveraging the Information Bottleneck principle to enhance privacy preservation alongside improving learning performance. For instance, Md Palash et al. [21] introduced a loss function grounded in the disentangled information bottleneck (DIB) framework, coupled with a mutual information (MI)-based model selection strategy, to mitigate the challenges of high communication overhead and data heterogeneity inherent in conventional FL settings. Beyond improving efficiency, such disentanglement indirectly contributes to privacy by restricting unnecessary information flow. Focusing more explicitly on IB's role in representation learning across domains, Li et al. [22] proposed an FL approach that employs the IB objective to encourage the learning of domain-invariant representations without necessitating direct data sharing among clients, thereby inherently safeguarding private client data. Moreover, their scale reweighting mechanism adjusts the influence of heterogeneous domain data scales on local objectives, further refining the extraction of privacy-preserving shared features. Yang et al. [23] advanced this direction with Federated Feature Distillation, which utilizes the IB principle to decouple data into performance-sensitive features —shared globally to combat data heterogeneity—and performance-robust features, which are retained locally to protect sensitive information. This selective sharing aligns with privacy objectives by minimizing exposure of less critical yet potentially sensitive data. Lastly, Diao et al. [24] proposed clustering clients by label distributions and training within clusters to handle extreme label skew. They introduced an IB-inspired stitching method that aggregates features from cluster-specific models.

Chapter 3 Preliminaries

3.1 Data Privacy Preservation

In our work, we focus on leveraging the Information Bottleneck framework to safeguard data privacy during the training process. Let the input data be denoted by X, which encapsulates both task-relevant information and potentially privacy-sensitive details. The target labels for the primary prediction task are represented by Y. To simultaneously enable privacy preservation and effective representation learning, we aim to extract a latent representation, denoted by Z. Specifically, given our emphasis on privacy protection, we seek to learn a privacy-preserving representation, denoted by Z^p, which explicitly filters out private or sensitive components present in the input. A key aspect of our approach lies in balancing the retention of sufficient information from X to ensure accurate prediction of Y against the suppression of redundant or sensitive details that could lead to privacy leakage. This trade-off is governed by a balancing parameter β. Moreover, in practical FL scenarios involving multiple clients k, each client may exhibit distinct privacy requirements. Consequently, identifying client-specific trade-off parameters β_k and learning corresponding privacy-preserving representations Z_k^p constitute the primary focus of our work.

3.2 Information Bottleneck Principle

Under the Markov chain $Y \to X \to Z$, the Information Bottleneck principle [9] aim to find a representation that maximizes the objective:

$$\min_{p(z|x)} I(X;Z) \, - \, \beta \, I(Z;Y) \, , \tag{3.1}$$

for given inputs denoted by X and outputs denoted by Y. Following Tishby's deep learning explanation [25], a deep learning procedure, we first target finding a representation Z that keeps the most information of Y. Once such Z is found, if the training is continued, the next goal is to find a way to compress X so that the information X that can reach Z is as little as possible. Given the two to be considered simultaneously, we have the IB theory, which maximizes $I(Z;Y)$ and minimizes $I(X;Z)$ at the same time, up to the balancing factor β.

Notably, by the definition of mutual information,

$$I(Z;Y) = H(Y) - H(Y \mid Z)\,, \tag{3.2}$$

where $H(Y)$ is the entropy of the target variable Y, representing its inherent uncertainty, and $H(Y \mid Z)$ is the remaining uncertainty about Y given the representation Z. If the data distribution and the task itself remain unchanged, then $H(Y)$ is a constant. In this case, maximizing $I(Z;Y)$ is equivalent to minimizing the conditional entropy $H(Y \mid Z)$. Therefore, Eq. 3.1 can be equivalently written as

$$\min_{p(z|x)} \; I(X;Z) \; + \; \beta\, H(Y \mid Z)\,. \tag{3.3}$$

Particularly, for classification tasks, the conditional entropy is commonly represented by the cross-entropy loss.

Regarding to data privacy preservation, one may think of the IB theory Indeed has the effect to keep as little information of X as possible once the learning of Y (from Z^p) is accomplished. That is, if we have already confirm the model effectiveness when predicting Y, we then think of if we can remove all other irrelevant information of X if it cannot help in predicting Y. In that case, we keep as much data privacy as possible. Following the rationale, we can design a data privacy mechanism by tuning the parameter β. Choosing a small β is equivalent to not paying attention to data privacy preservation, while choosing a large β is to encourage to keep the data privacy from leakage as much as possible.

Chapter 4 Methodology

4.1 IB-based Personalized Privacy Preservation in Federated Learning

As discussed in Section 3, inspired by the Information Bottleneck principle, we adopt an IB-based mechanism within the federated learning framework to enhance personalized data privacy preservation. In this framework, we consider diverse privacy preservation scenarios across different clients. The privacy preservation strategy can either be coordinated to achieve a global consensus among clients or be individually determined by each client according to its specific privacy requirements. Meanwhile, distinct from Eq. 3.1, our approach places a particular emphasis on data privacy protection by directly controlling the amount of information about the input X. Specifically, in our formulation, the trade-off coefficient β is applied to the $I(X;Z)$ term, allowing us to regulate how much information about X is retained in the learned representation Z_i^p. Thus, the objective can be written as:

$$\min_{Z} \ \beta_k \, I(X; Z_k^p) - I(Z_k^p; Y)\,, \quad \forall k\,, \tag{4.1}$$

For different client k, the information flow controlling parameter β_k now may become a variable that is a client-dependent parameter now. That is, a client k can decide its β_k, or up to the "guild" which has a certain degree of controlling power over each client, if the client has the motivation to join the guild, which is the federated learning framework. On the other hand, if a client k has a small amount of data, or the client wishes to reach the standard of k-anonymity, it may need to assign a relatively larger β_k in the formulation. Meanwhile, as shown in Section 3, Eq. 3.1 can be equivalently written as Eq. 3.3. Therefore, Eq. 4.1 can be rewritten as:

$$\min_{Z}\ \beta_k\, I(Z_k^p; X) + H(Y|Z_k^p)\,, \quad \forall k\,, \tag{4.2}$$

4.2 Framework Architecture

The methodology of this study incorporates IB into the local model, enabling feature compression during the training process to make it more difficult for adversaries to infer local data via Membership Inference Attacks. Most existing defense methods apply the highest level of protection uniformly across all clients, which, while ensuring privacy, often comes at the cost of model performance. In contrast, our proposed dynamic adjustment strategy determines the optimal compression rate for the data distribution of each individual client. Specifically, clients assessed as having higher privacy risks receive stronger compression and privacy protection, whereas clients with lower risks are subjected to moderate levels of defense in order to maintain overall model performance and avoid excessive degradation caused by overprotection. By leveraging the theoretical framework of IB, privacy-sensitive features that are more susceptible to inference attacks are compressed during training, while critical features relevant to the classification task are preserved to ensure model efficacy. The proposed framework is illustrated in Fig. 4.1.

We employ an encoder q_ϕ to estimate the mean μ and standard deviation σ from the features output by the previous layer. This allows us to map the features to a probabilistic distribution in the latent space, and, by performing random sampling from a Gaussian distribution, obtain a new representation z from the approximated prior distribution. As shown in Fig.4.2, we insert the IB layer immediately after the first Group Normalization layer, enabling the model to learn features that have undergone normalization. The resulting learned feature distribution is thus more stable, which contributes to improved model performance [26]. Taking the widely used convolutional neural network ResNet for image tasks as an example, the model typically consists of several layers as the feature extractor at the front, depending on its depth and parameter count, followed by a fully connected layer for the final classification task. Each layer is composed of two residual blocks. We insert the

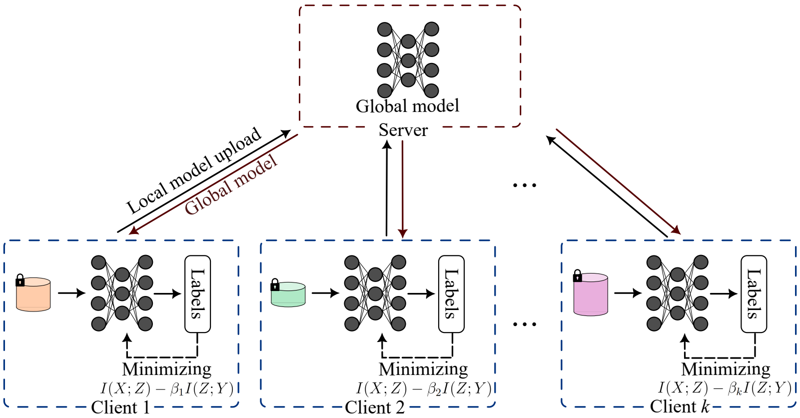


Figure 4.1: Framework of FedIB

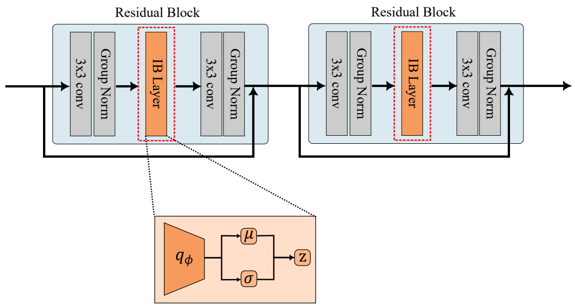


Figure 4.2: IB Layer Structure.

IB layer after the first Group Normalization within each of the two residual blocks in layer, allowing the model to learn from the most normalized features. It should be noted that we do not add IB layers to all layers of the network. Experimental results indicate that incorporating the IB layer into the shallowest layers most effectively leverages the privacy protection benefits of compression. Inserting IB layers into too many layers would require finely tuning the compression rate for each layer; otherwise, excessive compression of features could lead to a decline in model performance.The relevant experiments can be found in Subsection 5.4 for reference.

4.3 When IB Meets Privacy and Federated Learning

In the domain of FL, significant disparities in privacy risk arise due to the heterogeneous data distributions and diverse privacy requirements of individual clients. Specifically, certain clients may possess more sensitive or rare data, which, if not adequately protected through targeted privacy mechanisms, could be more easily identified by adversaries. However, applying the highest level of defense uniformly across all clients for the sake of overall security can lead to substantial degradation in the performance of the global model. Striking a balance between meeting diverse privacy demands and avoiding excessive defense-induced performance deterioration is a central issue addressed in this study. Moreover, the presence of non-IID local data distributions may further exacerbate the risk of clients being subjected to MIA.

4.4 MIA Risks under Non-IID Settings

When the local data of FL clients exhibits non-IID characteristics, the model tends to memorize unique local sample distributions, which significantly increases the success rate of MIA. According to our experimental results Fig. 4.3, we observe that after introducing the IB defense mechanism, clients with a higher degree of data non-IID require stronger information compression to maintain the same level of privacy protection. However, increasing the compression strength also leads to a notable decline in model performance. To address this issue, we design an adaptive compression framework that quantifies the local data imbalance using two metrics. This enables us to identify the most suitable compression strength for each client ' s data, thereby achieving an optimal balance between privacy protection and model utility.

We first employ Shannon entropy to measure the diversity of the local label distribution:

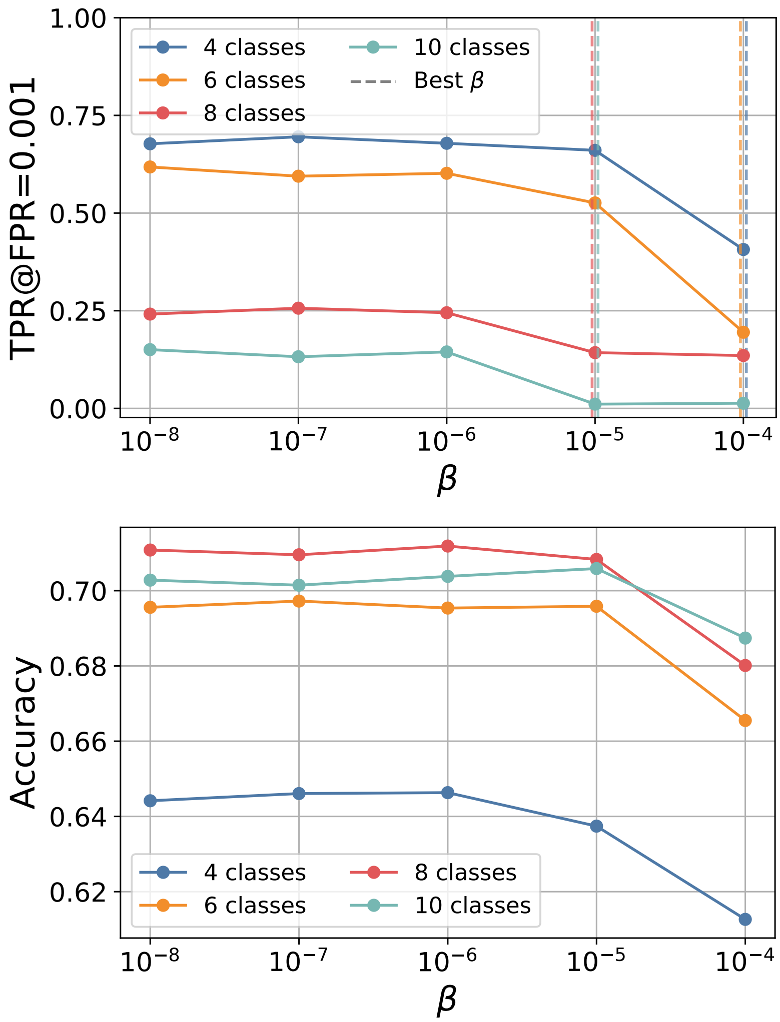


Figure 4.3: Comparison of attack success rates and model accuracy under varying numbers of client classes.

$$Entropy = 1 + \frac{1}{\log(n_c)} \sum_{i=1}^{n_c} p_{c_i} \log(p_{c_i}) \tag{4.3}$$

Here, n_c denotes the total number of classes, and p_{c_i} represents the proportion of samples belonging to class i. A larger value of this metric indicates a more imbalanced data distribution and, consequently, a higher level of privacy risk.

The second metric is the Imbalance Ratio (IR), defined as the ratio of the number of samples in the largest class to that in the smallest class, which is used to capture cases of extreme imbalance.

$$IR = 1 - \frac{n_c - 1}{n_c} \sum_{i=1}^{n_c} \frac{n_{c_i}}{n - n_{c_i}} \tag{4.4}$$

Here, n_{c_i} denotes the number of samples in class i, and n is the total number of samples. A larger value of this metric indicates that the data is more skewed toward a single or a few classes, which corresponds to a higher potential privacy risk.

Finally, based on the computed entropy or imbalance ratio (IR) values, we map these metrics to a predefined range of information bottleneck compression strengths for dynamic adjustment:

$$\beta = \beta_{\min} + (\beta_{\max} - \beta_{\min}) \times \text{Entropy or } IR \tag{4.5}$$

In the formulation, the maximum and minimum compression rates are set to $\beta_{\min} = 10^{-5}$ and $\beta_{\max} = 10^{-3}$, respectively. This allows the compression strength to be adaptively assigned based on each client ' s data risk score, enabling personalized privacy protection. As a result, it is possible to mitigate the risk of membership inference attacks (MIA) while preserving the performance of the global model.

4.5 Algorithm

The proposed method consists of two main stages. First, the imbalance of each client's local data is assessed by calculating both the entropy and imbalance ratio (IR) scores to quantitatively characterize the local data distribution. Next, these scores are mapped to a predefined compression range, as described in Eq. 4.5. The specific compression range is determined by the server based on prior experience, taking into account the characteristics of the training data and model for the given task.

After completing the local data assessment, each client obtains a personalized compression rate β_k tailored to its own data characteristics. During local training, the client applies its corresponding β_k and utilizes Eq. 4.7 to perform privacy-preserving feature compression. The resulting model, with private features suppressed, is then uploaded to the server for aggregation. As a result, even if an attacker gains access to the complete model, it becomes difficult to infer privacy-sensitive information, thereby enhancing overall privacy protection.We summarize the proposed algorithm as follows:

Algorithm 1 Personalized Beta Calculation

Require: Number of total clients K, maximum beta β_{max}; minimum beta β_{min}

1: **for** each client $k = 0$ to $K - 1$ **do** ▷ Determine the personalized compression rate β_k

2: Compute $Entropy_k$ over local data

3: Compute IR_k over local data

4: $\beta_k = \beta_{min} + (\beta_{max} - \beta_{min}) \cdot (Entropy_k \text{ or } IR_k)$

5: **end for**

6: **return** $\{\beta_k\}_{k=1}^{K}$

After the calculation of personalized compression rates, the federated learning training phase proceeds for T global rounds. In each round, a subset S_t consisting of r randomly selected clients from the total K clients is chosen to participate in the current training round and receive the broadcasted global model from the server. Each selected client then performs E local epochs of training on its local dataset of size N_k. During local training, each client utilizes its pre-computed personalized compression rate β_k to train its local model. Upon completion of local updates, the clients upload their updated models to the server. Once all participating clients have completed their uploads, the server aggregates the received models to form the updated global model for the next round. This process is repeated until T global rounds are completed.

Algorithm 2 FedIB Framework with Personalized Adaptive β

Require: Number of total clients K, number of selected clients per round r; global rounds T; local epochs E; learning rate η; local data size N_k

Server Execution:

Initialize global model θ^0
for $t = 0$ to $T-1$ **do**
  $\mathcal{S}_t \leftarrow$ Randomly sample r clients
  for each $k \in \mathcal{S}_t$ **do**
    $\theta_k^{t+1} \leftarrow$ **ClientUpdate**(θ^t)
  end for
  Aggregate: $\theta^{t+1} \leftarrow \frac{1}{N_{total}} \sum_{k \in \mathcal{S}_t} N_k \cdot \theta_k^{t+1}$, where $N_{total} = \sum_{k \in \mathcal{S}_t} N_k$
end for
return θ^T

ClientUpdate(θ^t)**:**

$\theta_k^{t,0} \leftarrow \theta^t$
for $e = 0$ to $E-1$ **do**
  $\theta_k^{t,e+1} \leftarrow \theta_k^{t,e} - \eta \nabla L(\theta_k^{t,e}, \beta_k)$ ▷ Compress the features using loss Eq.(4.7), with the personalized compression rate β_k
end for
return $\theta_k^{t,E}$

4.6 Learning Objective

As described in Subsection 4.1, Eq. 4.2 is designed to obtain a representation Z_k^p that simultaneously achieves personalized privacy protection and preserves model performance. Specifically, the term $I(Z_k^p; X)$ serves to constrain the personalized input information contained in Z_k^p, thereby enhancing privacy protection. To this end, we train an encoder $q_{\phi \sim \mathbb{P}}$, parameterized by ϕ and drawn from the distribution $\mathbb{P}$, to obtain the desired representation Z_k^p.

However, computing $I(Z_k^p; X)$ in deep learning models is highly challenging. To address this issue, we follow the upper-bound approximation method for $I(X; Z_k^p)$ proposed in [27], and use the Negative Log-Likelihood to approximate the conditional entropy $H(Z_k^p|X)$, as detailed below:

$$\begin{aligned}\mathcal{L}_{\text{VIB}} = & \underbrace{\mathbb{E}_{p(x,y)}\, \mathbb{E}_{q_\phi(z^p|x)}\Big[-\log p_\theta(y \mid z^p)\Big]}_{\text{Negative Log-Likelihood}} \\ & + \beta\ \underbrace{\text{KL}\Big[q_\phi(z^p \mid x) \,\|\, p(z^p)\Big]}_{\text{KL}}\end{aligned} \tag{4.6}$$

The encoder q_ϕ maps the input x to a stochastic latent variable z, and the amount of information about x retained in z^p is measured by the Kullback—Leibler (KL) divergence between q_ϕ and the prior distribution $p(z^p)$. The decoder p_θ then generates the corresponding output y conditioned on z^p.

Furthermore, since this study focuses on classification tasks, we employ Cross Entropy as a surrogate for the original Negative Log-Likelihood to represent the conditional entropy $H(Z_k^p|X)$. For each client k, our objective is to minimize the following personalized loss:

$$\mathcal{L} = \text{CE}\Big(y, p_\theta(y \mid z_k^p)\Big) + \beta_k\ \text{KL}\Big[q_{\boldsymbol{\phi}}(z_k^p \mid x) \,\|\, p(z^p)\Big] \tag{4.7}$$

Chapter 5 Experiments

5.1 Experimental Setup

5.1.1 Datasets and Models

In terms of experimental design, this study adopts two widely used public image datasets, CIFAR-10 and CIFAR-100, as the basis for evaluating the effectiveness of the proposed defense mechanism. The experiments are conducted under both IID and non-IID data distribution scenarios to comprehensively assess the generalizability and robustness of the proposed method.

For the IID data setting, we employ ResNet18 [28] as the baseline model and conduct experiments on both CIFAR-10 and CIFAR-100 datasets to comprehensively evaluate the proposed approach. In the non-IID scenario, ResNet18 is used as the backbone model and trained on CIFAR-100, where non-IID partitions are created using the Dirichlet distribution with varying concentration parameters α. We consider α values of 1.0, 10.0, and 100.0 to simulate different degrees of data heterogeneity, where $\alpha = 1.0$ represents the most imbalanced distribution and $\alpha = 100.0$ approximates the IID case.

5.1.2 Federated Setting

We adopt the classical federated learning algorithm FedAvg [1] for model training. In the IID experimental setting, the number of clients is set to 10, with a total of 100 global training epochs. Each FL client possesses 5,000 local data samples and uploads the model after training for one local epoch. For local training, we employ SGD as the optimizer with a learning rate of 0.001.

For the non-IID setting, we use the Dirichlet distribution with different α parameters to control the degree of non-IIDness. Similarly, we set 10 clients to participate in training and configure the global training epochs to 40. Each client uploads model parameters after 4 local training epochs.

5.1.3 MIAs

We employ three membership inference attack methods to evaluate the effectiveness of the proposed defense mechanism: Grad-Diff [11], Avg-Cosine [11], and FedMIA-Cosine [12]. The first two attacks exploit the model's prediction outputs, leveraging the overfitting characteristics for inference. In contrast, the latter three methods perform attacks based on the gradient information of the target model.

5.1.4 Defenses

We evaluate our proposed method against four existing baseline defenses. In particular, InstaHide [14] achieves protection through data augmentation, while the other approaches, including Differential Privacy (DP) [7], P^2-Protection [16], and FedDPA [20], attain defense by injecting noise or applying misleading perturbations to the uploaded gradients.

5.1.5 Evaluation Metric

We use the metrics AUC and TPR@FPR [12,29] to evaluate the extent of sample leakage under attack. Specifically, TPR@FPR refers to the True Positive Rate at a specified False Positive Rate in binary classification. This metric is particularly concerned with the TPR when the FPR is very low, thus providing a more effective reflection of the actual impact of the attack. In recent years, this metric has been widely adopted for evaluating MIA. Since the effectiveness of defenses depends on the degree of perturbation or noise added to the uploaded model by different defense parameters, the strength of the attack may be reduced, but this may also significantly impact model performance. Therefore, we use task accuracy to assess the effect of various defense mechanisms on model utility.

5.2 Comparison of MIA Defense Effectiveness

First, we compare the proposed method with several existing defense mechanisms on the CIFAR-10 dataset, evaluating their performance against three types of attacks. The detailed experimental results are presented in 5.1. The results demonstrate that our method consistently exhibits stable and significant defense capability across all attack scenarios. For the Grad-Diff, Avg-Cosine, and FedMIA-Cosine, both TPR and AUC metrics show substantial reductions, indicating a marked decrease in attack success rates. Although the defense effectiveness of our method is comparable to that of FedDPA under certain attacks, It is important to note that FedDPA results in a substantial reduction in model accuracy, with a decrease of approximately 0.324, which could severely impact real-world applications. In contrast, our approach maintains effective defense while only incurring approximately 0.027 performance drop, which is a significant improvement over existing methods.

The ability of our method to balance defense effectiveness and model performance primarily stems from compressing privacy-related features during training. This approach effectively removes features associated with privacy while retaining information highly relevant to the task objective. Additionally, this process helps eliminate irrelevant noise, thereby not only ensuring model privacy but also minimizing adverse effects on model performance.

The experimental results on the CIFAR-100 dataset are summarized in Table 5.2. Due to the increased difficulty of the CIFAR-100 task, the performance differences among various defense methods are more pronounced. Our approach demonstrates substantial defense effectiveness against all three attacks, achieving the lowest AUC scores in each case. Moreover, our approach maintains strong model accuracy, exhibiting no significant degradation and, in some cases, even achieving slight improvements. These results further validate that our method effectively

Table 5.1: Comparison of different defense methods on the CIFAR-10 dataset.

MIA Method		Defense Method					
		No Def.	Instahide	DP	P^2-Protection	FedDPA	Ours
Grad-Diff	TPR (↓)	0.193	0.211	0.120	0.194	0.066	**0.058**
	AUC (↓)	0.706	0.555	0.574	0.708	**0.500**	0.684
Avg-Cosine	TPR (↓)	0.080	0.035	0.077	0.047	**0.000**	0.052
	AUC (↓)	0.632	0.517	0.545	0.643	**0.513**	**0.513**
FedMIA-Cosine	TPR (↓)	0.184	0.014	0.094	0.141	**0.000**	0.011
	AUC (↓)	0.693	0.528	0.535	0.699	**0.551**	0.675
Accuracy (↑)		0.733	0.328	0.322	0.322	0.409	**0.706**

suppresses task-irrelevant noise through privacy compression while preserving both privacy and model utility.

5.3 Comparison of Model Performance under Non-IID Settings

We further compare the proposed method with other defense approaches under three different non-IID settings. The degree of data heterogeneity is controlled by partitioning the dataset with a Dirichlet distribution using concentration parameters α set to 1.0, 10.0, and 100.0, representing varying levels of non-IIDness.

Under the highly non-IID setting ($\alpha = 1$) as shown in Table 5.3, our method achieves TPR and AUC scores across all three attack types that, while not the highest, are comparable to those of other defense methods. This demonstrates that our approach remains effective in preserving privacy even under severe data heterogeneity. Notably, FedDPA yields an accuracy of only 0.064, indicating a significant performance drop with deeper models. In contrast, our method achieves much higher

Table 5.2: Comparison of different defense methods on the CIFAR-100 dataset.

MIA Method		**Defense Method**					
		No Def.	Instahide	DP	P^2-Protection	FedDPA	Ours
Grad-Diff	TPR (↓)	0.256	0.339	0.180	0.168	**0.044**	0.066
	AUC (↓)	0.639	0.612	0.524	0.575	0.549	**0.512**
Avg-Cosine	TPR (↓)	0.595	0.233	0.060	0.045	**0.013**	0.067
	AUC (↓)	0.881	0.588	0.586	0.662	0.586	**0.558**
FedMIA-Cosine	TPR (↓)	0.702	0.268	0.071	0.110	**0.029**	0.067
	AUC (↓)	0.911	0.601	**0.582**	0.712	0.586	0.586
Accuracy (↑)		0.419	0.106	0.038	0.012	0.107	**0.421**

Table 5.3: Comparison of model performance under Non-IID Ratio α=1.

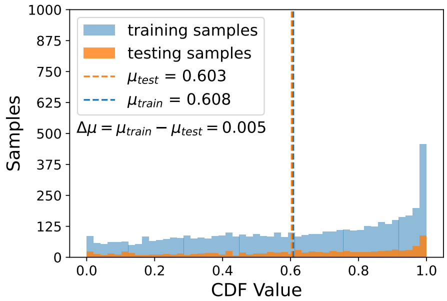
MIA Method		Defense Method						
		No Def.	Instahide	DP	P^2-Protect	FedDPA	Ours - Entropy	Ours - IR
Grad-Diff	TPR (↓)	0.075	0.054	0.039	0.072	**0.018**	0.055	0.037
	AUC (↓)	0.626	0.560	**0.516**	0.628	0.598	0.579	0.572
Avg-Cosine	TPR (↓)	0.340	0.169	0.153	0.040	**0.035**	0.082	0.086
	AUC (↓)	0.950	0.835	0.753	0.852	**0.740**	0.798	0.787
FedMIA-Cosine	TPR (↓)	0.608	0.412	0.418	**0.223**	0.279	0.327	0.294
	AUC (↓)	0.975	0.853	**0.766**	0.962	0.843	0.860	0.846
Accuracy (↑)		0.316	0.070	0.059	0.011	0.064	**0.283**	0.218

Table 5.4: Comparison of model performance under Non-IID Ratio α=10.

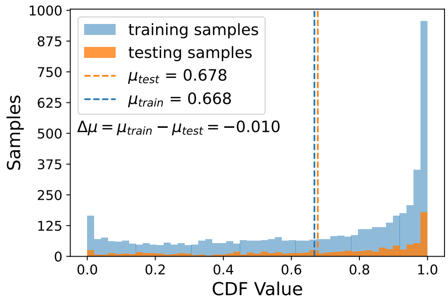
MIA Method		Defense Method						
		No Def.	Instahide	DP	P^2-Protect	FedDPA	Ours - Entropy	Ours - IR
Grad-Diff	TPR (↓)	0.057	0.057	0.039	0.063	**0.008**	0.037	0.033
	AUC (↓)	0.626	0.560	**0.502**	0.629	0.569	0.580	0.578
Avg-Cosine	TPR (↓)	0.337	0.062	0.118	0.018	**0.015**	0.021	0.022
	AUC (↓)	0.922	0.643	0.630	0.779	**0.556**	0.612	0.610
FedMIA-Cosine	TPR (↓)	0.571	0.141	0.139	0.269	**0.017**	0.082	0.066
	AUC (↓)	0.962	0.655	0.631	0.946	**0.590**	0.665	0.666
Accuracy (↑)		0.322	0.083	0.059	0.010	0.068	**0.316**	0.258

Table 5.5: Comparison of model performance under Non-IID Ratio α=100.

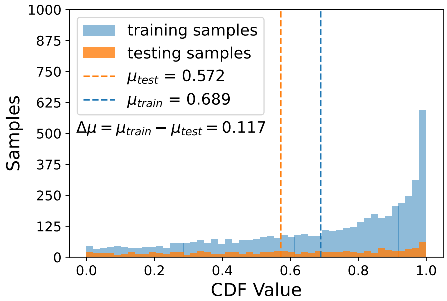
MIA Method		Defense Method						
		No Def.	Instahide	DP	P^2-Protect	FedDPA	Ours - Entropy	Ours - IR
Grad-Diff	TPR (↓)	0.075	0.066	0.058	0.069	**0.013**	0.037	0.033
	AUC (↓)	0.608	0.572	**0.521**	0.606	0.557	0.580	0.578
Avg-Cosine	TPR (↓)	0.347	0.012	0.040	0.035	**0.005**	0.021	0.022
	AUC (↓)	0.928	0.563	0.565	0.773	**0.509**	0.612	0.610
FedMIA-Cosine	TPR (↓)	0.578	0.033	0.038	0.214	**0.000**	0.082	0.066
	AUC (↓)	0.964	0.587	**0.569**	0.942	0.570	0.665	0.666
Accuracy (↑)		0.326	0.081	0.060	0.017	0.074	**0.316**	0.261



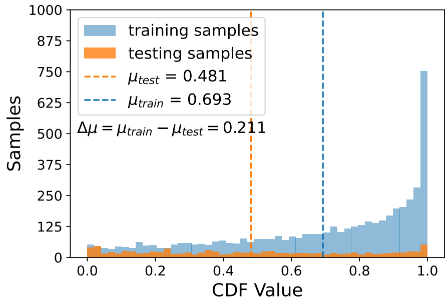
(a) IB in Layer1



(b) IB in Layer2



(c) IB in Layer3



(d) IB in Layer4

Figure 5.1: MIA for Member and Non-member Samples with IB Layer Inserted in Different Layers.

accuracies of 0.283 and 0.218 for the Entropy and IR metrics, respectively, clearly outperforming FedDPA and highlighting the robustness of our approach in complex model settings and highly heterogeneous data distributions. This result underscores the ability of our method to maintain a favorable balance between privacy protection and model utility.

Under the near-IID setting ($\alpha = 100$) as shown in Table 5.5, the accuracy of FedDPA increases only slightly to 0.074, which remains significantly lower than the accuracies of 0.316 and 0.261 achieved by our method. Both FedDPA and our method exhibit extremely low TPR and AUC scores across all MIA methods, This indicates that our approach is able to maintain both model performance and privacy protection.

In summary, our method consistently outperforms FedDPA in model accuracy across all non-IID conditions, with the most pronounced improvement observed under highly heterogeneous scenarios ($\alpha = 1$). At the same time, FedIB achieves TPR and AUC scores comparable to FedDPA across various MIA attack metrics, demonstrating that it can enhance model performance while maintaining a high level of privacy protection.

5.4 Effect of Layer-wise Feature Compression on Model Performance and Privacy

In this section, we discuss the impact of inserting IB layers at different network layers on privacy protection and model performance. MIA exploit differences in the model's outputs between training and non-training samples to distinguish member samples. By incorporating the IB during training, we encourage the model to learn only features relevant to classification, while compressing as much information related to the original data as possible. This approach reduces the model's behavioral differences between training and testing samples, thereby lowering the adversary's ability to distinguish membership.

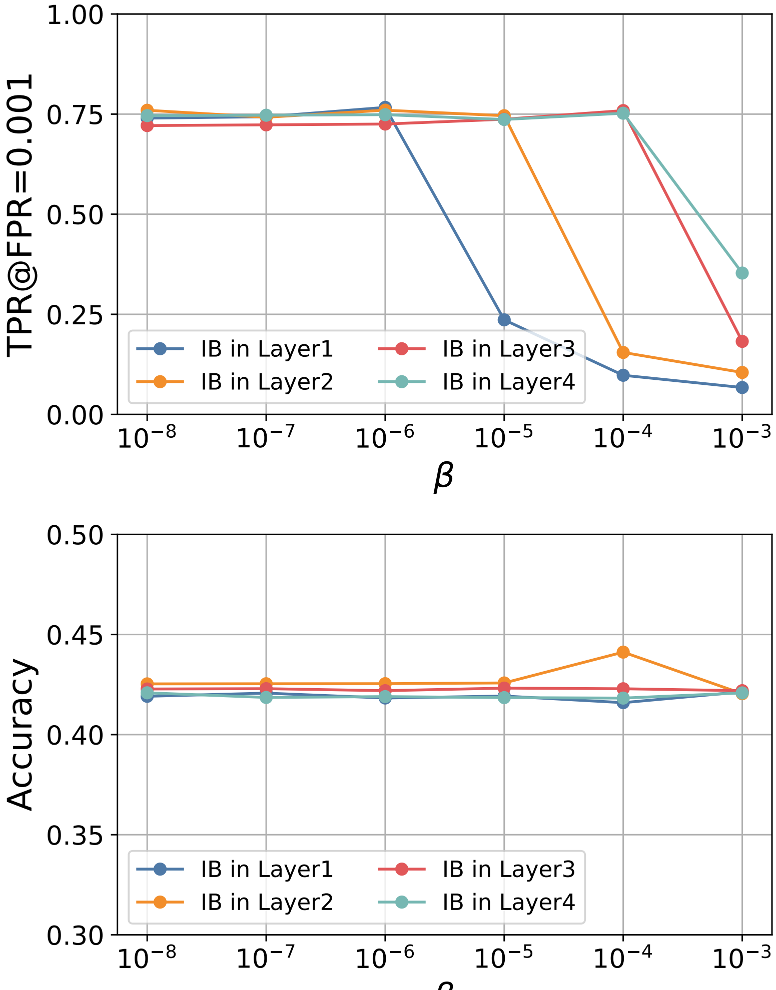


Figure 5.2: Defense effectiveness when inserting the IB at different position.

However, in the training process, each neural network layer captures different types of features. In image classification models, shallow layers typically extract coarse features such as contours or shapes, whereas deeper layers capture more refined features that are closely associated with classification tasks. Determining which layers should undergo feature compression to eliminate the gap between training and testing samples is a key challenge. Over-compressing all layers may hinder the model's ability to learn critical information, thus compromising its performance.

In this section, we use ResNet18 [28] as the backbone architecture, dividing the model sequentially into four layers, and inserting an IB layer after the first Group Normalization in every residual block of each layer. In this scenario, the compressed features are the richest and most closely related to the final classification task.

First, Fig. 5.1 illustrates the cumulative distribution function (CDF) scores for member and non-member samples when the IB layer is inserted at four different positions. Here, the CDF score serves as the basis for FedMIA in determining whether a particular sample is a training member. Specifically, FedMIA calculates the cosine similarity between the gradient of each sample and the model parameter update direction, and then maps the outputs of the target model to the distribution of outputs from shadow models to obtain the CDF scores for each sample, thereby identifying member samples. When the distribution gap between training and test samples is large, attackers can more easily distinguish member from non-member samples, leading to a higher risk of privacy leakage. Conversely, if the distributions are similar, the defense effectiveness is enhanced, making it more difficult for attackers to correctly identify member samples. In this experiment, we inserted IB layers at four different depths in the model and analyzed their effects on the CDF score distributions of training and test samples.

The experimental results show that when the IB layer is placed in shallower layers, the CDF distributions of training and test samples become more aligned. This phenomenon suggests that compressing features at shallow layers helps prevent the model from overfitting to specific sample information, promotes feature generalization, and thereby improves privacy protection.

Additionally, Fig. 5.2 further demonstrates the impact on TPR and accuracy when the IB layer is inserted at different layers. The experiments indicate that placing the IB layer in shallow layers with a compression rate of $\beta = 10^{-5}$ can already significantly enhance defense effectiveness. Moreover, as the compression parameter increases to $\beta = 10^{-3}$, the introduction of IB at shallower positions continues to exhibit better defense performance. It is worth noting that, according to the accuracy results, the compression operation does not have a significant negative impact on model performance, regardless of which layer the IB layer is inserted into.

Chapter 6 Conclusions

In summary, although existing enhanced defense mechanisms can effectively mitigate the risk of privacy leakage, they often incur significant losses in model performance. Moreover, current dynamic adjustment methods based on DP are frequently constrained by the scale of model parameters. To address these challenges, this study proposes an IB-based defense strategy that preserves privacy while minimizing the negative impact on model utility. Furthermore, we dynamically adjust the defense strength according to the local data distribution of each client, applying stronger protection only to high-risk clients. This approach achieves a better balance between privacy protection and overall model performance.

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